8. Solving the electric potential

1) The uniqueness theorem

① Green's first identity

Let $\vec{F} = \phi\,\nabla\psi$, $\quad \nabla\cdot\vec{F} = \nabla\phi\cdot\nabla\psi + \phi\,\nabla^2\psi$

$\because\ \int \nabla\cdot\vec{F}\,dV = \oint \vec{F}\cdot d\vec{a}$ (Gauss's theorem)

$\therefore\ \int (\nabla\phi\cdot\nabla\psi + \phi\,\nabla^2\psi)\,dV = \oint \phi\nabla\psi\cdot\hat{n}da$

$$\Longrightarrow \boxed{\int (\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi)\,dV = \oint \phi\frac{\partial\psi}{\partial n}\,da}$$

② The uniqueness theorem

The charge density distribution ρ in a region is given, while the potential ϕ satisfies $\nabla^2\phi = {}^{\rho}/_{\varepsilon_0}$; on the boundary of the region, either $\phi|_s$ or $\left.\frac{\partial\phi}{\partial n}\right|_s$ is specified. These conditions ensure that the solution to ϕ is unique.

Proof:

Assume: $\begin{cases} \nabla^2\phi_1 = {}^{\rho}/_{\varepsilon_0} \\ \phi_1|_s = f_s \text{ or } \left.\frac{\partial\phi_1}{\partial n}\right|_s = g_s \end{cases}$, $\quad \begin{cases} \nabla^2\phi_2 = {}^{\rho}/_{\varepsilon_0} \\ \phi_2|_s = f_s \text{ or } \left.\frac{\partial\phi_2}{\partial n}\right|_s = g_s \end{cases}$

(ϕ_1, ϕ_2 are two different solutions to the Poisson's equation with the same boundary conditions)

set $\Phi = \phi_1 - \phi_2$, $\quad$ now $\begin{cases} \nabla^2\Phi = 0 \quad \text{(Laplace's equation)} \\ \Phi|_s = 0 \quad \text{or} \quad \left.\frac{\partial\Phi}{\partial n}\right|_s = 0 \end{cases}$

Using Green's first identity and setting $\phi = \psi = \Phi$:

$$\Longrightarrow \int (\Phi\nabla^2\Phi + (\nabla\Phi)^2)\,dV = \oint \Phi\frac{\partial\Phi}{\partial n}\,da$$

Since $\nabla^2\Phi = 0$, $\quad \Longrightarrow \int_v (\nabla\Phi)^2 dV = \oint_s \Phi\frac{\partial\Phi}{\partial n}\,da$;

Since $\Phi|_s = 0$ or $\left.\frac{\partial\Phi}{\partial n}\right|_s = 0$, $\quad \Longrightarrow \int (\nabla\Phi)^2\,dV \equiv 0 \quad \Rightarrow \nabla\Phi = 0$

$\Rightarrow \nabla\phi_1 = \nabla\phi_2 \Rightarrow \vec{E}_1 - \vec{E}_2 = 0 \Rightarrow$ There can't be two different solutions for the electric field.

Note:

(i) There are several kinds of boundary conditions:

(a) Dirichlet: $\phi|_s$ is given.

(b) Neumann: $\left.\frac{\partial \Phi}{\partial n}\right|_s$ is given.

(c) $Modified\ Neumann$: $A\phi|_s + B\left.\frac{\partial \Phi}{\partial n}\right|_s$ is given (Robin).

(ii) A special type of Poisson's equation (the homogeneous Poisson's equation), $\nabla^2\Phi = 0$, is called the Laplace's equation.

(iii) Laplace's equation with Neumann boundary condition may have no solution:

The Neumann boundary condition specifies $\left.\frac{\partial \phi}{\partial n}\right|_S$, but

$$\oint_S \frac{\partial \phi}{\partial n}\, da = \oint_S (\nabla\phi)\cdot\hat{n}da = \oint_S (\nabla\phi)\cdot d\vec{a} = \int_V \nabla\cdot(\nabla\phi)dV = \int \nabla^2\phi\, dV \xrightarrow{Laplace} \equiv 0.$$

So, for any Neumann boundary condition that does not satisfy $\oint_S \frac{\partial \phi}{\partial n}\, da = 0$, there is no solution.

(iv) The uniqueness theorem guarantees that:

"As long as A SOLUTION is found, it is THE SOLUTION."

(v) The existence of the solution is difficult to prove. We only prove the uniqueness of the solution.

Example:

Prove the uniqueness of the solution to the following situation, in which $\phi|_S$ or $\left.\frac{\partial \phi}{\partial n}\right|_S$ is not directly specified: For a region with conductors, in which the charge amount on each conductor is known, while outside the conductors the distribution of charge is specified (as shown below), the solution to the potential ϕ is unique.

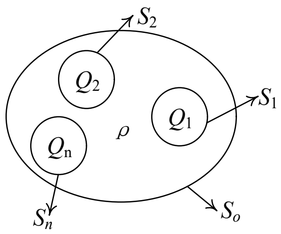

Figure. A region with multiple conductors.

Proof:

Assume there are two possible solutions, set $\Phi = \phi_1 - \phi_2$, they both satisfy the Poisson's equation

$\Rightarrow\ \nabla^2\Phi = 0$. Applying the Green's first identity:

$\int_V (\nabla\Phi)^2 dV = \sum_{i=0}^{n} \oint_{S_i} \Phi \frac{\partial\Phi}{\partial n}\, da$, since each conductor is an equipotential,

$\int_V (\nabla\Phi)^2 dV = \sum_{i=0}^{n} \Phi_i \oint_{S_i} \frac{\partial\Phi}{\partial n}\, da$, due to the B.C. of conductors, $= \sum_{i=0}^{n} \Phi_i \oint_{si} \left(-\frac{\sigma_1-\sigma_2}{\varepsilon_0}\right) da$

$= -\sum_{i=0}^{n} \frac{\Phi_i}{\varepsilon_0}(Q_{i1} - Q_{i2}) = 0$. The only possibility is $\nabla\Phi = 0$, meaning $\vec{E}_1 - \vec{E}_2 = 0$.

Note, here ϕ was solved for the region outside the conductors, so the boundaries of the conductors are treated as the boundary of the region.

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2) The method of images

Now let's look at one example which is difficult to solve without introducing the method of images.

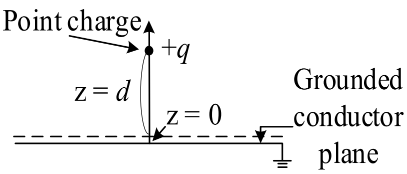

Figure. The configuration of a point charge above a grounded conductor plane.

In this problem, the entire z = 0 plane is a grounded conductor. A point charge is positioned at a distance d above the origin. Solve the potential ϕ in the region z > 0 (above the conductor plane).

Solution:

In the region z>0, the potential ϕ satisfies the Poisson's equation: $\nabla^2\phi = -\frac{q\delta^3(0,0,d)}{\varepsilon_0}$, which is subject to the boundary conditions $\begin{cases}\phi = 0, & at\ z = 0 \\ \phi \to 0, & \text{far from } (0,0,d) \text{ in the upper plane}\end{cases}$.

If the surface charge distribution σ on the $z = 0$ plane is known, we can directly compute the potential from the following integration:

$$\phi(\vec{r}\,) = \underbrace{\frac{1}{4\pi\varepsilon_0}\int_{z=0} \frac{\sigma(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|}\, da'}_{\text{surface contribution}} + \underbrace{\frac{1}{4\pi\varepsilon_0}\frac{q}{|\vec{r} - \vec{r}_d|}}_{\text{point charge contribution}}$$

Unfortunately, $\sigma(\vec{r}\,')$ is unknown, and we are stuck.

A clever way to get around is to bear in mind that as long as we can find another configuration which satisfies ① the same Poisson's equation in the region z > 0 (which means you don't introduce extra charge into this region but you can bring in charges elsewhere), and, ② the boundary conditions are unchanged (in this particular example, $\phi = 0$ for the z = 0 plane and $\phi \to 0$ far from the (0, 0, d) point in the upper plane), the solution to this new configuration will be THE SOLUTION in the z > 0 region.

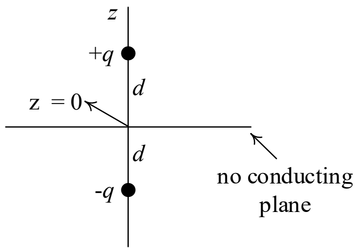

Figure. The configuration created using the method of images.

We construct the new configuration by introducing an image charge with opposite polarity, placed at (0, 0, – *d*), which is symmetrically positioned relative to the positive charge about the z = 0 plane. Note, in this new configuration, there is no conducting plane at z = 0. First, in the z > 0 region, no extra charge in introduced, so Poisson's equation in that region is unaltered; second, due to the symmetry of the new configuration, it is easy to verify that on the z = 0 plane $\phi = 0$, while far from (0, 0, *d*) in the upper plane, $\phi \to 0$. So the potential of the second configuration in region z > 0 is the solution to the original problem, explicitly:

$$\phi(x, y, z) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{x^2+y^2+(z-d)^2}} - \frac{1}{\sqrt{x^2+y^2+(z+d)^2}}\right].$$

Due to uniqueness of the solution, this solution is THE solution in the upper half plane.

Conclusions:

(i) The method of images directly solves Poisson's equation by a clever guess, the legitimacy of the solution is guaranteed by the uniqueness theorem.

(ii) Never put the mirror charge in the region of interest.

(iii) The overall field should satisfy the original boundary condition.

3) Separation of variables

The separation of variables method is a popular approach for solving Poisson's equation. The goal is to determine the potential ϕ in a region with given charge distribution ρ and boundary *S*. We illustrate this strategy using the Dirichlet boundary condition, which specifies ϕ on *S*.

$$\begin{cases} \nabla^2 \phi = -\dfrac{\rho}{\varepsilon_0} \\ \phi|_s = f \end{cases}$$

We split the solution into two parts,

$$\phi = \phi_p + \phi_L$$

where ϕ_p is the solution to Poisson′s equation in FREE SPACE, i.e.,

$$\phi_p = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho}{\mathscr{r}} dV'$$

and ϕ_L is the solution to Laplace′s equation:

$$\begin{cases} \nabla^2 \phi_L = 0. \\ \phi_L|_s = f - \phi_p\big|_s \end{cases}$$

It is straightforward to check $\phi = \phi_p + \phi_L$ satisfies the original Poisson's equation and boundary condition. Since the first part of solution, ϕ_p, can be found by a simple integration, solving the entire problem boils down to solving Laplace's equation, which can be done using separation of variables.

Note:

① If charge is distributed within the solution region, we use the above approach to transform the problem into one that involves solving Laplace's equation. Conversely, if charge is absent in the solution region, we directly solve Laplace's equation.

② To better visualize the properties of the solution to Laplace's equation, we first prove that the height of an elastic membrane stretched over an open surface (where the height of the open surface serves as the boundary condition) satisfies two-dimensional Laplace's equation. The force applied to a surface element of the membrane is illustrated in the following figure.

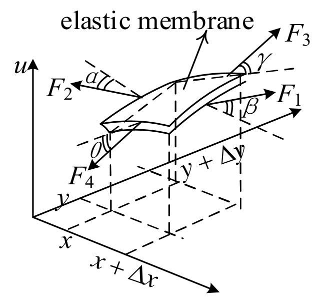

Figure. A surface element of a streched membrane and the forces applied on it.

Here *u* denotes the height (deflection in z), *T* is the uniform tension (force/unit length) in the membrane. Along the height direction, the force due to F_1, F_2, F_3, F_4 are $T\Delta y\ sin\,\beta$, $-T\Delta y\ sin\,\alpha$, $T\Delta x\ sin\,\gamma$, $-T\Delta x\ sin\,\theta$, respectively, where the values of the sine functions can be approximated using: $sin\,\beta \approx \partial_x u(x+\Delta x, y)$, $sin\,\alpha \approx \partial_x u(x, y)$, $sin\,\gamma \approx \partial_y u(x, y+\Delta y)$, $sin\,\theta \approx \partial_y u(x, y)$. In steady state, these forces cancel, resulting in

$$T\Delta y(\partial_x u(x+\Delta x, y) - \partial_x u(x, y)) + T\Delta x(\partial_y u(x, y+\Delta y) - \partial_y u(x, y)) = 0$$

Dividing the above equation by $T\Delta x\Delta y$, we get:

$$\frac{\partial_x u(x+\Delta x, y) - \partial_x u(x, y)}{\Delta x} + \frac{\partial_y u(x, y+\Delta y) - \partial_y u(x, y)}{\Delta y} = 0$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} u(x, y) + \frac{\partial^2}{\partial y^2} u(x, y) = 0$$

This is two-dimensional Laplace's equation: $\nabla^2 u(x, y) = 0$. As a result, the solution to Laplace's equation can be visualized as the shape of a stretched rubber sheet over an open surface, where the boundary condition determines its height. The following figure illustrates a representative example.

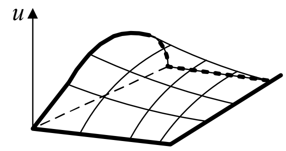

Figure. A representative illustration of the shape of a stretched elestic membrane.

A key property of the solution to Laplace's equation, $u(x, y)$, is that it has no local extrema within the solution region. This holds true in any number of dimensions.

Below, we illustrate the detailed procedure of using the separation of variables method to solve Laplace's equation in Cartesian and spherical coordinates through two examples.

Example 1. Calculate the electric potential in the cuboid region shown below, where all surfaces except the top one are grounded. The boundary at $z = c$ is maintained at a potential $V(x, y)$.

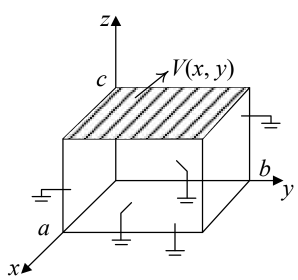

Figure. Configuration of the system in Example 1.

Solution:

Within the solution region, no charge is present; all charges reside on the six surfaces of the cuboid. Thus, the potential satisfies Laplace's equation with the following Dirichlet boundary conditions:

$$\begin{cases} \nabla^2\phi = 0 \\ \phi|_{x=0} = \phi_{x=a} = \phi_{y=0} = \phi_{y=b} = \phi_{z=0} = 0 \\ \phi_{z=c} = V(x,y) \end{cases}$$

Our strategy is to devise a solution in the form of

$$\phi(x,y,z) = X(x)Y(y)Z(z),$$

Here, X is a function of x only, etc. We try to find the expressions of X, Y, Z that satisfy Laplace's equation and the boundary conditions. Insert $\phi(x,y,z) = X(x)Y(y)Z(z)$ into $\nabla^2\phi = 0$, invoking $\nabla^2 \equiv (\partial_{xx} + \partial_{yy} + \partial_{zz})$, we obtain

$$X_{xx}YZ + XY_{yy}Z + XYZ_{zz} = 0$$

Here, for simplicity of notation, X represents $X(x)$, etc., and X_{xx} represents $\frac{\partial^2}{\partial x^2}X(x)$, etc. Since the uniqueness theorem guarantees a single solution, we can freely explore various solution approaches during the solving process. As long as we find one solution that satisfies Laplace's equation and the given boundary conditions, we can be certain it is the correct solution.

If $X_{xx} = -k_x^2X,\ Y_{yy} = -k_x^2Y,\ Z_{zz} = -k_x^2Z$, where $k_x^2,\ k_y^2,\ k_z^2$ are constants, and if $k_x^2 + k_y^2 + k_z^2 = 0$ is satisfied, we can straightforwardly check that $X_{xx}YZ + XY_{yy}Z + XYZ_{zz} = 0$ is automatically met.

The type of equation $X_{xx} = -k_x^2X$ has solutions in the form of $X(x) = Ae^{ik_xx} + Be^{-ik_xx}$, k_x can either be real or imaginary, depending on the boundary conditions. In the x direction, the boundary condition dictates that $X(0) = X(a) = 0$, so a general form satisfying this condition is

$$X(x) \sim \sin\frac{n\pi}{a}x,\ (n = 1, 2, 3, \ldots)$$

The same argument is true for the y component, so

$$Y(y) \sim \sin\frac{m\pi}{b}y,\ (m = 1, 2, 3, \ldots)$$

Since $k_x = \frac{n\pi}{a}$ and $k_y = \frac{m\pi}{b}$ are both real, k_z should be imaginary to satisfy $k_x^2 + k_y^2 + k_z^2 = 0$, quantitatively, $k_z = iK_{nm}$, where $K_{nm} = \sqrt{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2}$. To satisfy the boundary condition $Z(0) = 0$, $Z(c) = V(x, y)$, the function Z should be a hyperbolic sine function:

$$Z(z) \sim \sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2}\, z\right]. \quad (\text{Note: } \sinh(x) = \frac{e^x - e^{-x}}{2})$$

Put together, we now have reached a solution in the form of

$$\phi_{nm}(x,y,z) = \sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{b}y\right)\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,z\right].$$

Such a solution satisfies Laplace's equation and 5 out of 6 boundary conditions except the top boundary. We also notice that we have the freedom to choose *n* and *m*, and we haven't explored this dimension of flexibility yet. Moreover, due to the linearity of Laplace's equation and the already satisfied homogeneous boundary conditions ($\phi = 0$), the combination of terms with different *m* and *n* also satisfy Laplace's equation and the $\phi = 0$ boundary conditions:

$$\phi(x,y,z) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{b}y\right)\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,z\right],$$

where A_{nm} are a series of unknown coefficients. We force this form of solution to satisfy the boundary condition on the top surface:

$$V(x,y) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{b}y\right)\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,c\right].$$

To calculate the term A_{pq} (where *p* and *q* are two fixed positive integers), we take advantage of the orthogonality of the sinusoidal functions. Multiplying both sides by $\sin\left(\frac{p\pi}{a}x\right)\sin\left(\frac{q\pi}{b}y\right)$, and integrating *x* from 0 to *a*, integrating *y* from 0 to *b*,

$$\int_0^b\int_0^a V(x,y)\sin\left(\frac{p\pi}{a}x\right)\sin\left(\frac{q\pi}{b}y\right)dxdy$$

$$= \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{nm}\underbrace{\left(\int_0^a \sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{p\pi}{a}x\right)dx\right)}_{=\begin{cases}0, n\neq p\\ \frac{a}{2}, n=p\end{cases}}\underbrace{\left(\int_0^b \sin\left(\frac{m\pi}{b}y\right)\sin\left(\frac{q\pi}{b}y\right)dy\right)}_{=\begin{cases}0, m\neq q\\ \frac{b}{2}, m=q\end{cases}}\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,c\right]$$

$$\Rightarrow\quad A_{pq}\cdot\frac{ab}{4}\,\sinh\left[\sqrt{\left(\frac{p\pi}{a}\right)^2+\left(\frac{q\pi}{b}\right)^2}\,c\right] = \int_0^b\int_0^a V(x,y)\sin\left(\frac{p\pi}{a}x\right)\sin\left(\frac{q\pi}{b}y\right)dxdy$$

Now we readily obtain:

$$A_{nm} = \frac{4\int_0^b\int_0^a V(x,y)\sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{b}y\right)dxdy}{ab\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,c\right]}$$

which, when substituted into

$$\phi(x,y,z) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{b}y\right)\sinh\left[\sqrt{\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2}\,z\right],$$

is the solution to Laplace's equation in the solution zone satisfying all the boundary conditions.

Now we take one step further to learn how the method of separation of variables can be applied in spherical coordinates.

Example 2: A dielectric sphere is placed in a uniform electric field, the external field is pointing along the +z direction. The dielectric sphere has a permittivity of ε and a radius R. Calculate the potential everywhere.

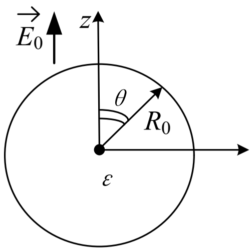

Figure. Configuration of the system in Example 2.

Solution:

The problem has symmetry in the azimuthal direction, i.e., the solution is independent of the angle φ. Moreover, just like in the last example, we guess a solution in the following form:

$$\phi\,(r,\theta) = \sum_{n=0}^{\infty}\left(a_n r^n + \frac{b_n}{r^{n+1}}\right) P_n\,(\cos\theta)$$

Here, $P_n(x)$ is the n′th order Legenere polynomial, in particular,

$$P_0(x) = 1\,, P_1(x) = x\,, P_n(1) = 1$$

The rationale to choose such a form is (1) it is a summation of infinite number of terms, (2) each term is a product of harmonic functions in their respective dimensions.

Assume that in the region external to the sphere, $\phi_1 = \sum_{n=0}^{\infty}\left(a_n r^n + \frac{b_n}{r^{n+1}}\right) P_n(\cos\theta)$, while for the region within the sphere, $\phi_2 = \sum_{n=0}^{\infty}\left(c_n r^n + \frac{d_n}{r^{n+1}}\right) P_n(\cos\theta)$. Next, we will need to determine the coefficients a_n, b_n, c_n, d_n, by applying the boundary conditions. However, in this problem the boundary conditions are not explicitly given as in the previous example.

First, consider the boundary at infinity, when $z \to +\infty$, the total field $\vec{E}_{tot} \to \vec{E}_0$, the potential as a function of r and θ is $\phi_1(z \to +\infty) = E_0 z = E_0 r\cos\theta$, see the following figure.

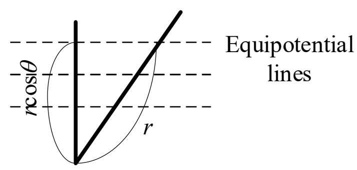

Figure. Illustration of equipotential lines at infinity.

Apparently, when $r \to +\infty$, only the a_1 term survives to satisfy the boundary condition at infinity, and a_n ($n \neq 1$) terms vanish. Thus,

$$\phi_1 = -E_0\, r\, P_1\,(cos\,\theta) + \sum \frac{b_n}{r^{n+1}}\, P_n\,(\cos\theta)$$

$$(a_1 = -E_0, a_0\,, a_2 \,\ldots\ldots a_n = 0)$$

Second, at the origin, ϕ_2 is finite, $d_n = 0\,, (n = 0\,, 1\,, 2\,\ldots)$.

Third, on the boundary of the sphere, the potential should be continuous, and $D_\perp$ is continuous (note, $D_\parallel$ is discontinuous because $P_\parallel$ is discontinuous). These conditions translate into:

$$\phi_1 = \phi_2,$$

$$\varepsilon_0 \frac{\partial \phi_1}{\partial r} = \varepsilon \frac{\partial \phi_2}{\partial r}.$$

The continuity of the potential:

$\phi_1 = \phi_2|_{r=R_0}$:

$$-E_0 R_0 P_1(\cos\theta) + \sum_{n=0}^{\infty} \frac{b_n}{{R_0}^{n+1}} P_n(\cos\theta) = \sum_{n=0}^{\infty} c_n R_0^n P_n(\cos\theta)$$

The continuity of $D_\perp$:

$\varepsilon_0 \dfrac{\partial \phi_1}{\partial r} = \varepsilon \dfrac{\partial \phi_2}{\partial r}\Big|_{r=R_0}$:

$$-E_0 P_1(\cos\theta) - \sum_{n=0}^{\infty} \frac{(n+1)b_n}{R_0^{n+2}} P_n(\cos\theta) = \frac{\varepsilon}{\varepsilon_0}\sum_{n=0}^{\infty} n\, c_n R_0^{n-1} P_n(\cos\theta)$$

All the terms in the above equations are functions of the Legendre polynomials, which need to match term-by-term when *n* takes different values.

For the $n \neq 1$ terms,

$$\begin{cases} \dfrac{b_n}{R_0^{n+1}} = c_n R_0^n \\ -\dfrac{(n+1)b_n}{R_0^{n+2}} = \dfrac{\varepsilon}{\varepsilon_0}\, n\, c_n R_0^{n-1} \end{cases}$$

If $b_n \neq 0, c_n \neq 0 \;\Rightarrow\; -\frac{1}{n+1} = \frac{\varepsilon_0}{\varepsilon}\cdot\frac{1}{n}$, which is impossible. As a result, $b_n = c_n = 0\,, n \neq 1$.

For the $n = 1$ term,

$$\begin{cases} -E_0 R_0 + \dfrac{b_n}{{R_0}^2} = c_n R_0^n \\ -E_0 - \dfrac{2b_1}{R_0^3} = \dfrac{\varepsilon}{\varepsilon_0} c_1 \end{cases}$$

Solving this set of equations, we get

$$b_1 = \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0 R_0^3\,, c_1 = -\frac{3\varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0.$$

The final result is

$$\begin{cases} \phi_1 = -E_0 r\cos\theta + \underbrace{\dfrac{\varepsilon-\varepsilon_0}{\varepsilon+2\varepsilon_0}\dfrac{E_0 R_0^3\cos\theta}{r^2}}_{\text{(Potential due to a dipole)}} \\ \phi_2 = \underbrace{-\dfrac{3\varepsilon_0}{\varepsilon+2\varepsilon_0}E_0 r\cos\theta}_{\text{(Uniform inside the sphere)}} \end{cases}$$

Note:

Below is the analysis of what happens at the boundary of the sphere. See the figure below.

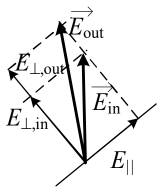

Figure. Electric field at the boundary of the dielectric sphere.

The horizontal component of the E field ($E_{||}$) is continuous. However, due to the existence of surface bound charge, the perpendicular component of the E field is discontinuous:

$$E_\perp^{out} - E_\perp^{in} = \sigma/\varepsilon_0\,,$$

On the upper part of the sphere, $\sigma > 0$ (because the external electric field points upward), so

$$E_\perp^{out} > E_\perp^{in}$$

As a result, electric field lines bend towards the surface normal direction, but cannot be perpendicular to the surface because $E_{||}$ is not zero.

The potential within the sphere is $\phi_2 = \frac{3}{\varepsilon_r+2}\,E_0 r\cos\theta$, from which we can compute the electric field:

$$\Rightarrow\ \vec{E}_{in} = \frac{3}{\varepsilon_r+2}\ \vec{E}_0$$

If $\varepsilon_r > 1$, $\vec{E}_{in} < \vec{E}_0$, which means the field inside the sphere is weaker than the external field. This is because the induced bound charge partially cancels the external field. However, incomplete cancellation results in the residue electric field, which is different from what happens to a sphere made of perfect conductor (i.e., complete cancellation of the internal field, see the example below). Now, let's study a highly relevant problem.

Example 3. Replace the dielectric sphere in the previous problem with a conductor sphere, calculate the electric potential everywhere.

Solution:

Since the approach is almost the same except the boundary conditions, we just briefly summarize the solution process below. We write the solution in the following form, for the potential external to the sphere and inside the sphere, respectively:

$$\phi_1(r,\theta) = \sum_{n=0}^{\infty}\left(a_n r^n + \frac{b_n}{r^{n+1}}\right)P_n(\cos\theta) \qquad \text{(external)}$$

$$\phi_2(r,\theta) = \sum_{n=0}^{\infty}\left(c_n r^n + \frac{d_n}{r^{n+1}}\right)P_n(\cos\theta) \qquad \text{(internal)}$$

Applying the following boundary conditions:

(1) At infinity, $\vec{E} = \vec{E}_0$, $\Rightarrow$ $\phi_1 \xrightarrow{\infty} -E_0 r P_1(\cos\theta)$ (i.e., $a_1 = -E_0$, a_0 , a_2 , …… $a_n = 0$);

(2) At the origin, ϕ_2 is finite, $\Rightarrow d_n = 0$, $n = 1,2,3 \ldots$

(3) At the boundary, $\phi_1 = \phi_2 =$ constant. It is convenient to set $\phi_1 = \phi_2 = 0$.

$$\phi_1 = -E_0 R_0 P_1 + \sum_{n=0}^{\infty}\left(\frac{b_n}{R_0^{n+1}}\right)P_n(\cos\theta) = 0$$

Note, P_n has $\cos\theta$ dependence, to satisfy the above equation, it requires

$$\begin{cases} b_0 , b_2 , b_3 \ldots = 0 \\ -E_0 R_0 P_1(\cos\theta) + \frac{b_1}{R_0{}^2} P_1(\cos\theta) = 0 \end{cases} \qquad \Rightarrow b_1 = E_0 R_0{}^3 , b_0 , b_2 , \ldots = 0.$$

So our final result is

$$\phi_1 = -E_0 r P_1 + \frac{E_0 R_0{}^3}{r^2} P_1(\cos\theta) , \ \phi_2 \equiv 0.$$

We can take one more step to compute the induced surface charge distribution:

$$\sigma(\theta) = -\varepsilon_0 \frac{\partial\phi_1}{\partial r}\bigg|_{r=R_0} = \varepsilon_0 E_0\left(1 + \frac{2R_0{}^3}{r^3}\right)\cos\theta\bigg|_{r=R_0} = 3\varepsilon_0 E_0 \cos\theta.$$

The electric field lines of a dielectric sphere and a metal sphere placed in a uniform external field are compared in the following figure.

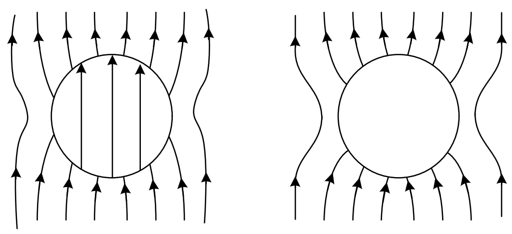

Figure. Electric field lines of a dielectric sphere (left) and a metal sphere (right).

9. Energy in electrostatics

Suppose $\vec{E}$ is the electric field due to a point charge. Now slowly (so slow that kinetic energy is negligible) bring a test charge Q from infinity to $\vec{r}$, where $\vec{\mathscr{r}}$ is the separation between these two charges. The electric field due to q is $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\mathscr{r}^2}\hat{\mathscr{r}}$. The force applied on the charge is $\vec{F} = -\frac{1}{4\pi\varepsilon_0}\frac{qQ}{\mathscr{r}^2}\hat{\mathscr{r}}$. The total energy stored is $\int_\infty^{\vec{\mathrm{r}}} \vec{F}\cdot d\vec{l} = Q\underbrace{\int_\infty^{\vec{r}}(-\vec{E})\cdot d\vec{l}}_{\varphi_q(\vec{\mathscr{r}})} = Q\phi_q(\vec{\mathscr{r}}) = \frac{Qq}{4\pi\varepsilon_0 \mathscr{r}}$.

If there are N point charges, the total energy required to assemble the system by bringing them one by one from infinity into their designated positions is

$$W = \frac{1}{4\pi\varepsilon_0}\sum_{i=2}^{N}\sum_{j<i}\frac{q_i q_j}{\mathscr{r}_{ij}}.$$

See the following table for the values of i and j in the above summation:

i	j
2	1
3	1, 2
4	1, 2, 3
…	…
N	1, 2, 3,…, N−1

We can prove that the above summation is equivalent to

$$W = \frac{1}{2}\cdot\frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\sum_{j\neq i}^{N}\frac{q_i q_j}{\mathscr{r}_{ij}}$$

Which is equivalent to $\frac{1}{2}\sum_{i=1}^{N} q_i \underbrace{\left(\sum_{i\neq j}^{N}\frac{1}{4\pi\varepsilon_0}\frac{q_j}{\mathscr{r}_{ij}}\right)}_{\substack{\text{potential at position } i \\ \text{due to OTHER charges}}}$. For continuous charge distribution, the result can be generalized into

$$W = \frac{1}{2}\int \rho(\vec{r})\,\phi(\vec{r})dV$$

$$= \frac{\varepsilon_0}{2}\int(\nabla\cdot\vec{E})\,\phi dV = \frac{\varepsilon_0}{2}\int(-\vec{E}\cdot\nabla\phi dV) + \frac{\varepsilon_0}{2}\int\nabla\cdot(\vec{E}\phi)dV$$

$$= \frac{\varepsilon_0}{2}\int\vec{E}\cdot(-\nabla\phi)dV + \frac{\varepsilon_0}{2}\underbrace{\oint(\phi\vec{E})\cdot d\vec{a}}_{\to 0,\ \text{since } (\frac{1}{r^2}\cdot\frac{1}{r}\cdot r^2\to 0)}$$

$$= \frac{\varepsilon_0}{2}\int\vec{E}^2 dV$$

Note:

① (I): $W = \frac{1}{2}\sum_{i=1}^{N} q_i\,(\vec{r}_i)\quad \underbrace{\phi(\vec{r}_i)}_{\text{due to OTHER charges.}}$ (for discrete charge distribution)

(II): $W = \frac{1}{2}\int \rho\,(\vec{r})\phi(\vec{r})dV$ (for continuous charge distribution)

(III): $W = \frac{\varepsilon_0}{2}\int \vec{E}^2 dV$ (general form)

② (III) tells us energy is stored in the field, and > 0

(II) sometimes is easier to compute (Why? We only integrate $\rho \neq 0$ region).

(I) can be negative → does not consider energy of the point charges themselves.

Energy of a point charge due to (III) is:

$$W = \frac{\varepsilon_0}{2}\cdot\int\left(\frac{q}{4\pi\varepsilon_0 r^2}\right)^2 dV = \frac{\varepsilon_0 q^2}{2\cdot(4\pi\varepsilon_0)^2}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty}\frac{1}{r^4}\cdot r^2 \sin\vartheta\, dr d\theta d\phi$$

$$= \frac{q^2}{8\pi\varepsilon_0}\int_0^{\infty}\frac{1}{r^2}\,dr \to \infty$$

This is an embarrassing result telling us that the classical theory predicts infinite energy for a single point charge. Without a more sophisticated theory to solve this problem, let's use (I) when the problem involves point charges. Otherwise, use (II) & (III) for calculation

③ Energy of a system with dielectric material can be calculated from

$$W = \frac{1}{2}\int \rho_f\,\phi dV\,,$$

This is because the free charge is what we can control and bring into the field. The dielectric material's response will modify the potential, which is already included in the term ϕ. Since $\rho_f = \nabla\cdot\vec{D}$, the above equation can be written as

$$\frac{1}{2}\int(\nabla\cdot D)\phi dV = \frac{1}{2}\int(-D)\cdot\nabla\phi dV + \frac{1}{2}\int\nabla\cdot(\vec{D}\phi)dV$$

$$= \frac{1}{2}\int\vec{D}\cdot\vec{E}dV$$

The general form of energy density, $\frac{\vec{D}\cdot\vec{E}}{2}$, accounts for both the energy stored in the field and in the polarized dielectric material (polarizing the materials takes great amount of work).

④ Energy does not obey the superposition principle, since

$$W_{tot} = \frac{\varepsilon_0}{2}\int E^2 dV = \frac{\varepsilon_0}{2}\int {E_1}^2 + {E_2}^2 + 2\vec{E}_1\cdot\vec{E}_2\, dV \neq W_1 + W_2.$$